

QUADRATIC EQUATIONS ...

SUB TOPICS

$$a^4 + a^2 + 1 = (a^2 - a + 1)(a^2 + a + 1)$$

- Relation b/w roots and coefficients
- Representation of Graphs
- Conditions for common roots
- Range of rational functions, $\mathbb{Q}$
- Theory of Equations
- Transformations of Quadratic Equations.
- Conditions for roots to be
 - $\Rightarrow$ Real.
 - $\Rightarrow$ Imaginary
 - $\Rightarrow$ Positive
 - $\Rightarrow$ Negative.
 - $\Rightarrow$ Of Opposite Sign
 - $\Rightarrow$ Both roots less than K
 - $\Rightarrow$ Both roots greater than K .
 - $\Rightarrow$ One roots less than K Other greater than K .
 - $\Rightarrow$ Both roots lie b/w $K_1 K_2$
 - $\Rightarrow$ Exactly one root lie b/w $K_1 K_2$
- Miscellaneous form:
 - $\Rightarrow$ Reciprocal form.
 - $\Rightarrow$ Differentiation Method.

"Completing Squaring Method" ...

FORMULAS $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

- Roots of Quadratic Equation,

$$\Rightarrow \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Q.E. whose roots are α, β $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

- In inequality

Odd Powers $\rightarrow$ change

Even Powers $\rightarrow$ constant

- $\alpha + \beta = -\frac{b}{a}$

$$\alpha\beta = \frac{c}{a}$$

$$|\alpha - \beta| = \frac{\sqrt{D}}{|a|} \quad (\because \text{Difference of roots})$$

Condition for common roots :

$$ax^2 + bx + c = 0$$

$$dx^2 + ex + f = 0$$

- $(af - cd)^2 = (bd - ae)(ce - fb)$

- Both roots are common

$$\boxed{\frac{a}{d} = \frac{b}{e} = \frac{c}{f}}$$

Range of Q.E.:

$$a > 0$$

$$\text{Range } \left\{ \left[\frac{-D}{4a}, \infty \right) \right\}$$

$$a < 0$$

$$\text{Range } \left\{ (-\infty, \frac{-D}{4a}] \right\}$$

* Range of Rational functions:

Case-1 : $\frac{\text{Linear}}{\text{Linear}}$

- Equival to y (Simplify)
- Get x in terms y

- Find Range.

Case - II $\frac{\text{Linear}}{\text{Quadratic}} - \frac{\text{Quadratic}}{\text{Quadratic}}$

$\Rightarrow$ $\frac{\text{Linear}}{\text{Quadratic}}$

- Equivalent to y
- Get Q.E
- Find discriminant
- Solve inequality
- Get Range by solve it

$$(a+b+c+d)^2 = a^2 + b^2 + c^2 + d^2 + 2(ab+ac+ad+bc+bd+cd)$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2 - ab - bc - ca)$$

$\Rightarrow$ $\frac{\text{Quadratic}}{\text{Quadratic}}$

- Only if coeff of x^2 is same
- Exp Num or denominator $x \pm y$
- Simple to form say $(1 \pm \frac{6}{x})$
- Now Find Min Max of Q.E.
- Find Range of it.

Theory of Equations:

- Cubic Equation whose roots are given:

$$x^3 - S_1 x^2 + S_2 x - S_3 = 0$$

$$ax^3 + bx^2 + cx + d \rightarrow (\alpha \beta \gamma)$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha \beta \gamma = -\frac{d}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$ax^4 + bx^3 + cx^2 + dx + e = 0 \rightarrow (\alpha \beta \gamma \delta)$$

$$\alpha + \beta + \gamma + \delta = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\delta + \delta\alpha + \alpha\gamma + \beta\delta = \frac{c}{a}$$

$$\alpha\beta\gamma\delta = \frac{e}{a}$$

$$\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \delta\alpha\beta = -\frac{d}{a}$$

Polynomial of n^{th} degree

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

$$\alpha_1 + \alpha_2 + \dots + \alpha_n = S_1 = -\frac{a_{n-1}}{a_n}$$

$$\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \dots = S_2$$

$$= +\frac{a_{n-2}}{a_n}$$

$$S_n = \alpha_1 \alpha_2 \dots \alpha_n$$

$$= (-1)^n \frac{a_0}{a_n}$$

Transformations:

- i. $\alpha + K, \beta + K \quad f(x - K)$
- ii. $\alpha - K, \beta - K \quad f(x + K)$
- iii. $K\alpha, K\beta \quad f\left(\frac{x}{K}\right)$
- iv. $\frac{\alpha}{K}, \frac{\beta}{K} \quad f(Kx)$
- v. $\alpha^2, \beta^2 \quad f(\sqrt{x})$
- vi. $\alpha^3, \beta^3 \quad f(\sqrt[3]{x})$
- * * * vii. $a\alpha + b, a\beta + b \quad f\left(\frac{x-b}{a}\right)$

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CUBIC equation whose zeros
are 1, 2, 3
: $x^3 - 6x^2 + 11x - 6 = 0$
roots are 1, 2, 3."
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Condition for roots to be

- i. Real $D \geq 0$
- ii. Imaginary $D < 0$
- iii. Positive $D \geq 0, -\frac{b}{a} > 0, \frac{c}{a} > 0$
- iv. Negative $D \geq 0, -\frac{b}{a} < 0, \frac{c}{a} > 0$
- v. of Opp Sign $D > 0, \frac{c}{a} < 0$
- vi. Both roots less than $K, D \geq 0, af(K) > 0, -\frac{b}{a} < 2K$
- vii. Both roots greater than $K, D \geq 0, af(K) > 0, -\frac{b}{a} > 2K$
- viii. $\alpha \rightarrow$ less than $K, D > 0, af(x) < 0$
 $\beta \rightarrow$ greater than K
- ix. Both lie b/w $K_1, K_2 : D \geq 0$
 $af(K_1) > 0, af(K_2) > 0$
* * * $-\frac{b}{a} > 2K_1, -\frac{b}{a} < 2K_2$
- x. $\alpha \rightarrow$ less than K_1
 $\beta \rightarrow$ greater than $K_2 : D > 0$
 $(K_1 < K_2) \quad af(K_1) < 0, af(K_2) < 0$
- xi. Exactly one root lie b/w $K_1, K_2 : D > 0, f(K_1)f(K_2) < 0$

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→ Miscellaneous forms

→ Finding roots (real using differentiation)

Range of $ax^2 + bx + c = 0 \Rightarrow \left\{ \frac{c - \frac{b^2}{4a}}, \frac{c + \frac{b^2}{4a}} \right\}$

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"FORMULAS"

$$\text{i. } (a+b+c)^2 = a^2 + b^2 + c^2 + 2abc(a+b+c)$$

$$\text{ii. } (a+b+c)^3 = a^3 + b^3 + c^3 - 3(ab+bc+ca)(a+b+c)$$
